



2024

Mathematics Extension 2 Task 3

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 1 hour
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- In Questions 6-8, show relevant mathematical reasoning and/or calculations

Total marks:
37

Section I – 5 marks (pages 2 – 3)

- Attempt Questions 1 – 5
- Allow about 8 minutes for this section.

Section II – 32 marks (pages 4 – 6)

- Attempt Questions 6 – 8
- Allow about 52 minutes for this section.

Section I

5 marks

Attempt Questions 1 – 5

Allow about 8 minutes for this section.

Use the multiple-choice answer page in the writing booklet for Questions 1 – 5.

- 1 A particle moves in a straight line with a displacement of x . When $t = 0$ the acceleration is $3x^2$, velocity is $-\sqrt{2}$ and displacement is 1. Which of the following is the correct equation for x as a function of t ?

A. $x = -\frac{2}{(t + \sqrt{2})^2}$

B. $x = -\frac{2}{(t - \sqrt{2})^2}$

C. $x = \frac{2}{(t + \sqrt{2})^2}$

D. $x = \frac{2}{(t - \sqrt{2})^2}$

- 2 The spheres $x^2 + y^2 + z^2 = 9$ and $x^2 + (y - 4)^2 + z^2 = 4$ intersect. What are the coordinates of the centre and radius of the circle in which they intersect?

A. Centre $\left(0, \frac{21}{8}, 0\right)$, radius $\frac{3\sqrt{15}}{8}$

B. Centre $\left(0, \frac{16}{9}, 0\right)$, radius $\frac{3\sqrt{15}}{8}$

C. Centre $\left(0, \frac{21}{8}, 0\right)$, radius $\frac{5\sqrt{15}}{8}$

D. Centre $\left(0, \frac{23}{8}, 0\right)$, radius $\frac{5\sqrt{15}}{8}$

- 3 A variable force acts on a particle causing it to move in a straight line. At time t seconds, its velocity v metres per second and position x metres from the origin are such that $v = e^x \cos x$.

Which of the following is an expression for the acceleration of the particle?

- A. $-e^x \sin x$
- B. $e^{2x} \cos x (\cos x - \sin x)$
- C. $e^x (\sin x + \cos x)$
- D. $2e^{2x} (\cos^2 x)$

- 4 A particle is moving in a straight line in simple harmonic motion. The displacement of the particle is given by $x = 6 \cos^2 t - 1$, where x is the displacement in metres and t is the time in seconds.

Where is the centre of motion?

- A. $x = -1$
- B. $x = 0$
- C. $x = 1$
- D. $x = 2$

- 5 The lines $\vec{r}_1 = \begin{bmatrix} 1 \\ -6 \\ 4 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 2 \\ a \end{bmatrix}$ and $\vec{r}_2 = \mu \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$, where λ and μ are scalar parameters and a is constant, are skew. Which of the following is NOT correct?

- A. a can equal -2
- B. a can equal -1
- C. a can equal 1
- D. a can equal 2

End of Section I

Section II

32 marks

Attempt Questions 6 – 8

Allow about 52 minutes for this section.

Instructions

- Answer the questions in the appropriate page of your answer booklet.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing paper is available. If you use extra paper, write your name, and your teacher's name on the paper and clearly indicate which question you are answering.

Question 6 (9 marks) Start on the appropriate page of your answer booklet

- a) Let A and B be the points on the line $\vec{r} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix}$ corresponding to $\lambda = 1$

and $\lambda = -1$ respectively. Let P be the point $(1, 0, 2)$.

- i) Find the projection of $\vec{AP}$ onto $\vec{AB}$. 3

- ii) Hence find the perpendicular distance d from P to the line AB . 2

- b) A particle is performing Simple Harmonic Motion in a straight line. At time t seconds it has displacement x metres from a fixed point O on the line, velocity $v \text{ ms}^{-1}$ given by

$$v^2 = 32 + 8x - 4x^2 \text{ and acceleration } a \text{ ms}^{-2}.$$

- i) Find an expression for a in terms of x . 1

- ii) Find the centre and amplitude of the motion. 2

- iii) Find the maximum speed of the particle. 1

Question 7 (12 marks) Start on the appropriate page of your answer booklet

- a) Relative to a fixed point at the origin, the points B , C and D are defined respectively by the position vectors $\underline{b} = \underline{i} - \underline{j} + 2\underline{k}$, $\underline{c} = 2\underline{i} - \underline{j} + \underline{k}$ and $\underline{d} = a\underline{i} - 2\underline{j}$, where a is a real constant.

Given that the magnitude of angle BCD is $\frac{\pi}{3}$, find the value of a . 4

- b) A line has the Cartesian equation $\frac{x+2}{4} = \frac{y-4}{5} = -\frac{z+1}{3}$. Find the Cartesian equation of a parallel line passing through the point with position vector $-2\underline{i} - 4\underline{j} + 5\underline{k}$. 4

- c) A sphere has a centre at $(3, -3, 4)$ and its radius is 6 units.

A line has equation $\underline{r} = \begin{pmatrix} 1 \\ 5 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix}$.

- i) Write down the vector equation of the sphere. 1
- ii) Determine whether the line is a tangent to the sphere, clearly justifying your conclusion. 3

Question 8 (11 marks) Start on the appropriate page of your answer booklet

- a) From an origin O , a parachutist of mass m kg falls from rest vertically downwards. The forces acting on the parachutist are his weight and a force due to air resistance of magnitude mkv^2 newtons, where k is a constant and $v \text{ ms}^{-1}$ is the parachutist's velocity. Let x be the parachutist's displacement in metres below O .

i) Show that the parachutist's equation of motion is $\ddot{x} = g - kv^2$ where g is the acceleration due to gravity. 1

ii) Use integration to show that $v^2 = \frac{g}{k}(1 - e^{-2kv})$. 2

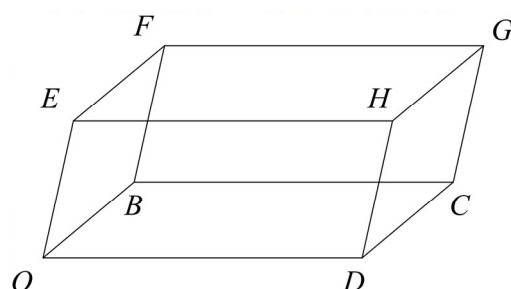
When the parachutist's velocity is $5g \text{ ms}^{-1}$, he opens his parachute. The motion is now modelled by assuming that the magnitude of the force due to air resistance instantaneously changes to $\frac{mgv}{10}$ newtons. The time from the parachute opening is t seconds.

iii) Use integration to show that $v = 10 + 5(g - 2)e^{\frac{-gt}{10}}$. 3

iv) It takes T seconds for the parachutist's velocity to decrease to $2g \text{ ms}^{-1}$. 1

Show that $T = \frac{10}{g} \ln \left[\frac{5(g - 2)}{2(g - 5)} \right]$.

- b) A parallelepiped is an oblique prism that has a parallelogram cross-section. It has three pairs of parallel and congruent faces.



$OBCDEFGH$ is a parallelepiped with $\underline{b} = \overrightarrow{OB}$, $\underline{d} = \overrightarrow{OD}$ and $\underline{e} = \overrightarrow{OE}$.

i) Express $\overrightarrow{OG}$ in terms of $\underline{b}$, $\underline{d}$ and $\underline{e}$. 1

ii) Prove that $|\overrightarrow{OG}|^2 + |\overrightarrow{DF}|^2 + |\overrightarrow{BH}|^2 + |\overrightarrow{CE}|^2 = 4(|\underline{b}|^2 + |\underline{d}|^2 + |\underline{e}|^2)$. 3

End of Paper



BAULKHAM HILLS HIGH SCHOOL
YEAR 12 H3 TEST 2024
MATHEMATICS EXTENSION 2

MARKING COVER SHEET

FINAL MARK

NESA #: _____ Teacher: _____

		Outcomes		
Qn	Parts	Vectors	Mechanics	Total
	Multiple Choice	2,5	1,3,4	
		/2	/3	/5
7	a	/5		/9
	b		/4	
8	a	/4		/12
	b	/4		
	c	/4		
9	a		/7	/11
	b	/4		
	Total	/23	/14	/37

Extension 2 Task 3 2024 solutions

Multiple choice

1. when $t=0$, both A and B have acceleration of -1 .

Consider C: $x = 2(t+\sqrt{2})^{-2}$

$$\frac{dx}{dt} = -4(t+\sqrt{2})^{-3}$$

$$v = \frac{-4}{(t+\sqrt{2})^3}$$

when $t=0$, $v = \frac{-4}{2\sqrt{2}}$

$$v = -\sqrt{2} \text{ m s}^{-1}$$

$\therefore$ C

or

$$a = 3x^{-\frac{3}{2}}$$

$$v \frac{dv}{dx} = 3x^{-\frac{3}{2}}$$

$$\int v dv = \int 3x^{-\frac{3}{2}} dx$$

$$\left[\frac{v^2}{2} \right]_{-\sqrt{2}}^v = \left[x^{-\frac{1}{2}} \right]_1^x$$

$$\frac{v^2}{2} - 1 = x^{-\frac{1}{2}} - 1$$

$$\frac{v^2}{2} = x^{-\frac{1}{2}}$$

$$v^2 = 2x^{-\frac{1}{2}}$$

when $x=1$, $v = -\sqrt{2}$

$$\therefore v = -\sqrt{2} \sqrt{x^{-\frac{1}{2}}}$$

$$\frac{dx}{dt} = -\sqrt{2} x^{-\frac{1}{2}}$$

$$\int x^{-\frac{1}{2}} dx = -\sqrt{2} \int dt$$

$$-2 \left[x^{\frac{1}{2}} \right]_1^x = -\sqrt{2} \left[t \right]_0^t$$

$$\frac{1}{\sqrt{x}} - 1 = \frac{1}{\sqrt{2}} t$$

$$\frac{1}{\sqrt{x}} = \frac{t}{\sqrt{2}} + \frac{\sqrt{2}}{\sqrt{2}}$$

$$\frac{1}{\sqrt{x}} = \frac{\sqrt{2} + t}{\sqrt{2}}$$

$$\sqrt{x} = \frac{\sqrt{2}}{\sqrt{2} + t}$$

$$x = \frac{2}{(t+\sqrt{2})^2}$$

$$\begin{aligned} \textcircled{2} \quad & x^2 + y^2 + z^2 = 9 \\ & x^2 + (y-4)^2 + z^2 = 4 \\ \hline & 8y - 16 = 5 \\ & 8y = 21 \\ & y = \frac{21}{8} \end{aligned}$$

$$\begin{aligned} \text{when } y = \frac{21}{8} \quad & x^2 + y^2 + z^2 = 9 \\ & x^2 + z^2 = \frac{135}{64} \end{aligned}$$

$\therefore$ centre is $(0, \frac{21}{8}, 0)$ and radius is $\frac{3\sqrt{5}}{8}$

$\therefore A$

$$\begin{aligned} \textcircled{3} \quad & v = e^{\lambda} \cos \lambda \\ a &= v \frac{dv}{d\lambda} \\ a &= e^{\lambda} \cos \lambda (\cos \lambda e^{\lambda} - e^{\lambda} \sin \lambda) \\ a &= e^{2\lambda} \cos \lambda (\cos \lambda - \sin \lambda) \\ \therefore & \textcircled{B} \end{aligned}$$

$$\begin{aligned} \textcircled{4} \quad & x = 6 \cos^2 t - 1 \\ & \cos 2\theta = 2 \cos^2 \theta - 1 \\ & x = 3(\cos 2t + 1) - 1 \\ & x = 3 \cos 2t + 2 \\ \therefore & \text{Centre of motion is } x = 2. \\ \therefore & \textcircled{D} \end{aligned}$$

$$\begin{aligned} \textcircled{5} \quad & \text{For lines to meet,} \\ & \text{(equating } x \text{ and } y) \quad \begin{array}{l} \textcircled{1} \times 2 \\ \textcircled{2} - \textcircled{3} \end{array} \end{aligned}$$

$$\begin{array}{rcl} 1 + \lambda & = & \mu \quad \textcircled{1} \\ -6 + 2\lambda & = & -2\mu \quad \textcircled{2} \\ \underline{2 + 2\lambda} & = & \underline{2\mu} \quad \textcircled{3} \\ -8 & = & -4\mu \\ \therefore \mu & = & 2 \end{array}$$

$$1 + \lambda = 2$$

$$\lambda = 1$$

$$\therefore \lambda = 1 \text{ \& } u = 2.$$

For lines to meet

$$4 + \lambda a = u$$

$$4 + a = 2$$

$$a = -2$$

$\therefore$ for lines to be skew $a \neq -2$

$\therefore$ (A)

6 a i) Point A: $\lambda = 1$

$$A = (5, -1, 0)$$

Point B: $\lambda = -1$ $B = (-1, 3, -2)$

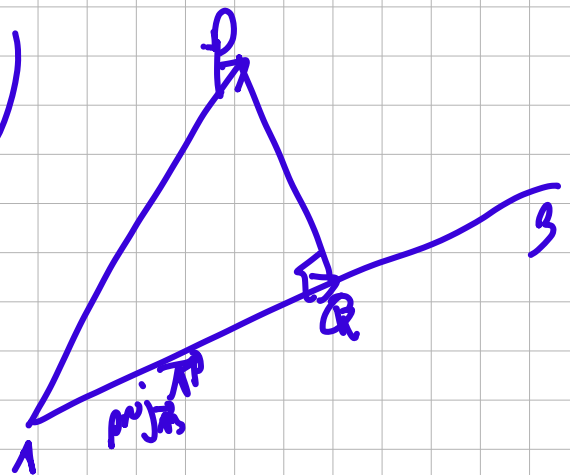
$$\vec{AP} = \begin{pmatrix} -4 \\ 1 \\ 2 \end{pmatrix} \quad \vec{AB} = \begin{pmatrix} -6 \\ 4 \\ -2 \end{pmatrix}$$

$$\text{proj}_{\vec{AB}} \vec{AP} = \frac{\vec{AP} \cdot \vec{AB}}{|\vec{AB}|^2} \vec{AB}$$

$$= \frac{24 + 4 - 4}{(\sqrt{56})^2} \begin{pmatrix} -6 \\ 4 \\ -2 \end{pmatrix}$$

$$= \frac{24}{56} \times 2 \begin{pmatrix} -3 \\ 2 \\ -1 \end{pmatrix}$$

$$= \frac{6}{7} \begin{pmatrix} -3 \\ 2 \\ -1 \end{pmatrix}$$



② correct solution

① finds $\vec{AB}$ or $\vec{AP}$

6 a ii) Perp distance = $|\vec{AP} - \text{proj}_{\vec{AB}} \vec{AP}|$

$$= \left| \begin{pmatrix} -4 \\ 1 \\ 2 \end{pmatrix} - \begin{pmatrix} -\frac{18}{7} \\ \frac{12}{7} \\ -\frac{6}{7} \end{pmatrix} \right|$$

② correct solution

① finds expression for perpendicular distance

$$= \sqrt{\frac{75}{7}}$$

$$= \frac{5\sqrt{3}}{\sqrt{7}} \text{ or } \frac{5\sqrt{21}}{7} \text{ units}$$

6 b) i) $a = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$

(1) correct solution

$$a = \frac{d}{dx} \left(\frac{1}{2} (32 + 8x - 4x^2) \right)$$

$$= \frac{d}{dx} (16 + 4x - 2x^2)$$

$$\therefore \ddot{x} = 4 - 4x$$

6 b ii) $\ddot{x} = -4(x-1)$

(2) correct solution

$\therefore$ centre of motion is $x=1$

(1) finds centre of motion or amplitude

$$v^2 = 32 + 8x - 4x^2$$

when $v=0$ $0 = 8 + 2x - x^2$

$$x^2 - 2x - 8 = 0$$

$$(x-4)(x+2) = 0$$

$$x = -2 \text{ or } x = 4$$

$$\text{Amplitude} = \frac{4 - (-2)}{2}$$

$$= 3$$

6 b iii) Max speed occurs at centre of motion

(1) correct solution

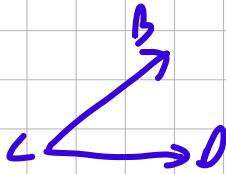
when $x=1$, $v^2 = 32 + 8 - 4$

$$v^2 = 36$$

$$\text{max speed} = 6 \text{ ms}^{-1}$$

$$7 a) \quad \underline{b} = \underline{i} - \underline{j} + 2\underline{k} \quad \underline{c} = 2\underline{i} - \underline{j} + \underline{k} \quad \underline{d} = a\underline{i} - 2\underline{j}$$

$$\angle BCD = \frac{\pi}{3}$$



$$\vec{CB} = \vec{OB} - \vec{OC}$$

$$\vec{CB} = -\underline{i} + \underline{k}$$

$$\vec{CD} = \vec{OD} - \vec{OC}$$

$$\vec{CD} = (a-2)\underline{i} - \underline{j} - \underline{k}$$

$$\cos \angle BCD = \frac{\vec{CB} \cdot \vec{CD}}{|\vec{CB}| |\vec{CD}|}$$

$$\cos \frac{\pi}{3} = \frac{-a+2-1}{\sqrt{2} \sqrt{(a-2)^2 + 1 + 1}}$$

$$\frac{1}{2} = \frac{a-1}{\sqrt{2} \sqrt{a^2 - 4a + 6}}$$

$$\sqrt{2} \sqrt{a^2 - 4a + 6} = 2(a-1)$$

$$2(a^2 - 4a + 6) = 4(a-1)^2$$

$$a^2 - 4a + 6 = 2(a^2 - 2a + 1)$$

$$a^2 - 4a + 6 = 2a^2 - 4a + 2$$

$$4 = a^2$$

$$a = \pm 2$$

④ correct solution

③ obtains expression in a

② substitutes into $a \cdot b = |a||b|\cos\theta$

① finds $\vec{CB}$ or $\vec{CD}$

$$7b) \quad \frac{x}{4} + \frac{1}{2} = \frac{4-y}{-5} = \frac{z+1}{-3}$$

$$\therefore \frac{x}{4} + \frac{1}{2} = 1$$

$$x+2=4\lambda$$

$$x = -2+4\lambda$$

$$\frac{4-y}{-5} = \lambda$$

$$4-y = -5\lambda$$

$$y = 4+5\lambda$$

$$\therefore \text{Line is } \vec{r} = \begin{pmatrix} -2 \\ 4 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 5 \\ -3 \end{pmatrix}$$

Line passing through $\begin{pmatrix} -2 \\ 4 \\ -1 \end{pmatrix}$ is

$$\vec{r} = \begin{pmatrix} -2 \\ 4 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 5 \\ -3 \end{pmatrix}$$

$$x = -2 + 4\lambda$$

$$y = -4 + 5\lambda$$

$$z = 5 - 3\lambda$$

$$4\lambda = x+2$$

$$5\lambda = y+4$$

$$3\lambda = -z+5$$

$$\lambda = \frac{x+2}{4}$$

$$\lambda = \frac{y+4}{5}$$

$$\lambda = \frac{-z+5}{3}$$

$$\therefore \text{Cartesian form } \frac{x+2}{4} = \frac{y+4}{5} = \frac{z-5}{-3}$$

(4) correct solution

(3) finds vector form of line

(2) finds direction vector

(1) finds a parametric

equation for x, y or z

$$\lambda = \frac{z+1}{-3}$$

$$-3\lambda = z+1$$

$$z = -1-3\lambda$$

7 < i) let z be a point on the sphere (1) correct equation

$$\therefore \left| z - \begin{pmatrix} 3 \\ -3 \\ 4 \end{pmatrix} \right| = 6 \text{ is the equation of the sphere}$$

7 < ii) $x = 1 + 2\lambda$

$$y = 5 - \lambda$$

$$z = 4 - \lambda$$

$$\therefore (1 + 2\lambda - 3)^2 + (5 - \lambda + 3)^2 + (4 - \lambda - 4)^2 = 36$$

$$(2\lambda - 2)^2 + (8 - \lambda)^2 + \lambda^2 = 36$$

$$4\lambda^2 - 8\lambda + 4 + 64 - 16\lambda + \lambda^2 + \lambda^2 = 36$$

$$6\lambda^2 - 24\lambda + 68 = 36$$

$$6\lambda^2 - 24\lambda + 32 = 0$$

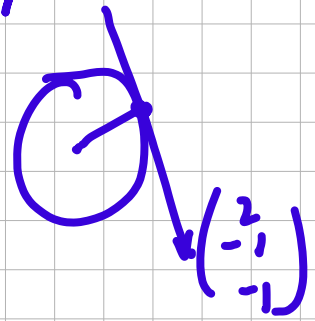
$$3\lambda^2 - 12\lambda + 16 = 0$$

$$\lambda = \frac{12 \pm \sqrt{144 - 4 \times 3 \times 16}}{2 \times 3}$$

$$= 12 \pm \sqrt{-48}$$

no real solution

$\therefore$ Line does not intersect the sphere.

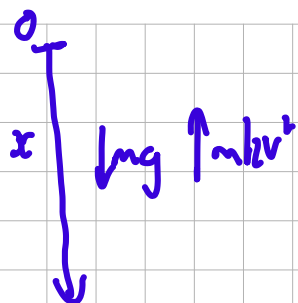


(3) correct solution

(2) solves/finds discriminant

(1) forms equation in λ

8a i)



$$F = ma$$

$$m\ddot{x} = mg - mkv^2$$

$$\ddot{x} = g - kv^2$$

① correct solution

8a ii)

$$v \frac{dv}{dx} = g - kv^2$$

$$\int_0^v \frac{v dv}{g - kv^2} = \int_0^x dx$$

$$[x]_0^x = \frac{-1}{2k} \int_0^v \frac{-2kv dv}{g - kv^2}$$

$$x - 0 = \frac{-1}{2k} \left[\ln |g - kv^2| \right]_0^v$$

$$-2kx = \ln |g - kv^2| - \ln |g|$$

$$-2kx = \ln \left| \frac{g - kv^2}{g} \right|$$

$$e^{-2kx} = \frac{g - kv^2}{g}$$

$$ge^{-2kx} = g - kv^2$$

$$kv^2 = g(1 - e^{-2kx})$$

$$v^2 = \frac{g}{k}(1 - e^{-2kx})$$

② correct solution

① correct primitive

8a iii)

$$m\ddot{x} = mg - \frac{mgv}{10}$$

$$\ddot{x} = g - \frac{gv}{10}$$

$$\frac{dv}{dt} = \frac{10g - gv}{10}$$

$$\frac{1}{g} \int_{5g}^v \frac{10 dv}{10 - v} = \int_0^t dt$$

$$\frac{10}{-g} \int_{5g}^v \frac{-1}{10 - v} dv = \left[t \right]_0^t$$

$$-\frac{10}{g} \left[\ln |10 - v| \right]_{5g}^v = t - 0$$

$$-\frac{g}{10} t = \ln |10 - v| - \ln |10 - 5g|$$

$$-\frac{g}{10} t = \ln \left| \frac{10 - v}{10 - 5g} \right| \quad (1)$$

$$e^{-\frac{gt}{10}} = \frac{10 - v}{10 - 5g}$$

$$5e^{-\frac{gt}{10}}(2 - g) = 10 - v$$

$$v = 10 - 5e^{-\frac{gt}{10}}(2 - g)$$

$$\therefore v = 10 + 5(g - 2)e^{-\frac{gt}{10}} \quad \text{as required}$$

8a iv) When $t = T, v = 2g$

$$\text{From (1) above } -\frac{gT}{10} = \ln \left| \frac{10 - 2g}{10 - 5g} \right|$$

$$T = -\frac{10}{g} \ln \left| \frac{2(5 - g)}{5(2 - g)} \right|$$

$$T = \frac{10}{g} \ln \left| \frac{5(2 - g)}{2(5 - g)} \right|$$

$$\therefore T = \frac{10}{g} \ln \left[\frac{5(g - 2)}{2(g - 5)} \right]$$

③ correct solution

② Integrates correctly

① correct integral

① correct proof

$$86 \text{ i) } \vec{OC} = \vec{OB} + \vec{BC} + \vec{CE}$$

$$\vec{OC} = \vec{b} + \vec{OD} + \vec{DE}$$

$$\vec{OC} = \vec{b} + \vec{d} + \vec{e}$$

$$\vec{OD} = \vec{BC}$$

$$\vec{CA} = \vec{DE}$$

① correct expression
opposite sides of parallelogram are equal and parallel

ii) Similarly,

$$\vec{OF} = \vec{b} + \vec{e} - \vec{d}$$

$$\vec{BH} = \vec{d} - \vec{b} + \vec{e}$$

$$\vec{CE} = -\vec{d} - \vec{b} + \vec{e}$$

③ correct proof

② expresses sum as dot products

① finds expressions for $\vec{OF}$, $\vec{BH}$ and $\vec{CE}$

$$|\vec{OC}|^2 + |\vec{OF}|^2 + |\vec{BH}|^2 + |\vec{CE}|^2$$

$$= (\vec{b} + \vec{d} + \vec{e}) \cdot (\vec{b} + \vec{d} + \vec{e}) + (\vec{b} + \vec{e} - \vec{d}) \cdot (\vec{b} + \vec{e} - \vec{d}) + (\vec{d} - \vec{b} + \vec{e}) \cdot (\vec{d} - \vec{b} + \vec{e}) + (-\vec{d} - \vec{b} + \vec{e}) \cdot (-\vec{d} - \vec{b} + \vec{e})$$

$$= \vec{b} \cdot \vec{b} + \vec{d} \cdot \vec{d} + \vec{e} \cdot \vec{e} + \cancel{\vec{b} \cdot \vec{d}} + \cancel{\vec{b} \cdot \vec{e}} + \cancel{\vec{d} \cdot \vec{b}} + \cancel{\vec{d} \cdot \vec{e}} + \cancel{\vec{e} \cdot \vec{b}} + \cancel{\vec{e} \cdot \vec{d}}$$

$$+ \vec{b} \cdot \vec{b} + \vec{e} \cdot \vec{e} + \vec{d} \cdot \vec{d} + \cancel{2\vec{b} \cdot \vec{e}} - \cancel{2\vec{b} \cdot \vec{d}} - \cancel{2\vec{d} \cdot \vec{e}}$$

$$+ \vec{d} \cdot \vec{d} - \cancel{2\vec{b} \cdot \vec{d}} + \cancel{2\vec{d} \cdot \vec{e}} - \cancel{2\vec{b} \cdot \vec{e}} + \vec{b} \cdot \vec{b} + \vec{e} \cdot \vec{e}$$

$$+ \vec{d} \cdot \vec{d} + \vec{b} \cdot \vec{b} + \vec{e} \cdot \vec{e} + \cancel{2\vec{b} \cdot \vec{d}} - \cancel{2\vec{b} \cdot \vec{e}} - \cancel{2\vec{d} \cdot \vec{e}}$$

$$= 4|\vec{b}|^2 + 4|\vec{d}|^2 + 4|\vec{e}|^2$$

$$= 4(|\vec{b}|^2 + |\vec{d}|^2 + |\vec{e}|^2)$$